

Kaden Seto

[Website](#) | kaden.seto@mail.utoronto.ca | [Linkedin](#) | [Github](#)

EDUCATION

University of Toronto

Bachelor of Applied Science & Engineering (BASc) in Engineering Science + PEY Co-op

Toronto, ON

Sep 2024 - 2029

TECHNICAL SKILLS

Languages: Python, C#, C/C++, JavaScript/TypeScript, Java, HTML/CSS, MATLAB

Developer Tools: Git, Visual Studio Code, Unity, Node.js, Android Studio

Libraries: Unity ML-Agents, PyTorch, Gymnasium & SB3, Keras, NumPy, pandas, OpenCV, scikit-learn

EXPERIENCE

University of Toronto Machine Intelligence Student Team | [Website](#)

Toronto, ON

Vice President, Academics (May 2025 - present)

Nov 2024 - present

- Management of the Academics department and initiatives, encompassing **12 directors and leads** across **5 Academic workshops, tournaments, and programs** (in collaboration with industry partners such as **IBM, AMD, Riot Games**) supporting **3000+ students** (North America's largest undergraduate ML/AI team)

Reinforcement Learning Team Lead (Nov 2024 - Mar 2025) | [AI² Website](#) | [Github](#)

- Directed a developer team to design UTMIST AI², **Canada's largest RL tournament**, for **400+ participants**
- Developed a custom **multi-agent RL self-play training process** that supports custom **PyTorch** and **SB3** agent implementations and a custom platform-fighting **game environment** using **Pygame** and **Pymunk**
- **First-authored** a paper presented and submitted at **CUCAI (Canada's largest undergrad AI conference)**

BMO Financial Group

Toronto, ON

Software Developer Intern

May 2025 - Aug 2025

- Developed scripts for automating environment syncing and AWS CDK Stack deployment, **reducing process time by 2+ hours**

PROJECTS

Continuum | [Github](#)

Oct 2025 - present

- Researching & developing a dimension-agnostic deep RL framework in **PyTorch** for **physics-informed reinforcement learning**, combining **SB3 model-free RL algorithms** (e.g. PPO, SAC), **Neural ODEs**, and **Autoencoders (AEs)** to build a custom neural network architecture for solving physics-based Gym environments.
- Achieving **200-300%** better reward performance compared to regular RL algorithms in MuJoCo environments.

NeuroStrike | [Github](#)

May 2025 - Aug 2025

- Engineered a RL framework for **training martial arts AI agents** to spar in a **custom-made 3D environment** simulation using **self-play recurrent PPO (LSTM + PPO)**.
- Engineered hyperparameters, architectures, reward functions, and observation spaces to optimize policy behaviour for optimal sampling of movement and attack actions.

Aegis | [Github](#)

Jan 2025 - Apr 2025

- Trained a **custom object detection model** using YOLOv11 architecture, achieving a **mAP50-95 of 84.6%**, average **50ms inference time**, and detections up to **3.2m** on a phone camera for a **Cyclist Detection + Arduino LED safety system** to prevent right-hooking incidents.
- Explored **super-resolution GAN** models, implementing **Swift-SRGAN in PyTorch** to enable real-time super-resolution for enhanced detection.

QuantNet | [Github](#)

Oct 2024 - Feb 2025

- Implemented a hybrid **LSTM & MLP neural network framework from scratch** with **0 external dependencies** (i.e. methods such as **back-propagation** and **Adam** made from scratch) for investment analysis
- Implemented a **feature engineering framework from scratch** by **researching key investment analysis indicators** to enhance model quality and performance (e.g. features such as RSI, ADX)

OtakuNet | [Github](#)

Aug 2024 - Sep 2024

- Developed a **content-based recommender system** to recommend animes based on user-rated genres, building a custom **dual-architecture neural network framework** from scratch (using only **NumPy**) and a custom **data engineering & preprocessing framework** for the task (e.g. features such as Bayesian ratings, normalization)